# Assignment 4

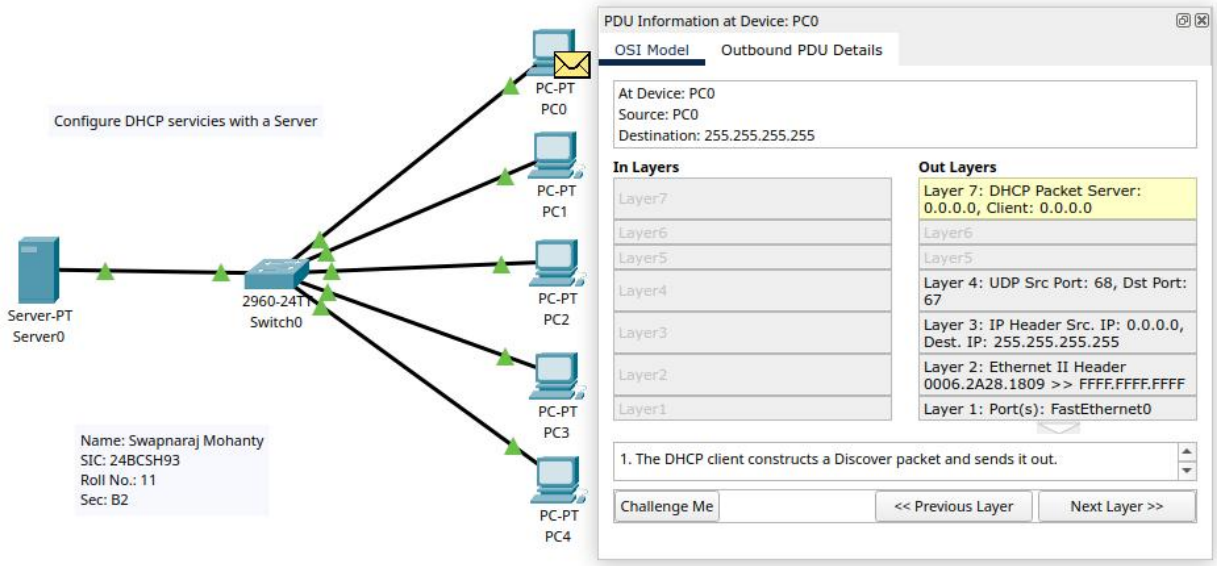
**Q1. Configure DHCP Services with a Server.**

1. Create one Server-PT, one Cisco 2960 switch, and five PCs in Cisco Packet Tracer.
2. Connect the server and all PCs to the switch using Copper Straight-Through cables.
3. Click Server0 → Desktop → IP Configuration and assign the static IP address 192.168.1.2, subnet mask 255.255.255.0, and default gateway 192.168.1.1.
4. Click Server0 → Services → DHCP, turn the DHCP service On, and configure the following settings:
   - Pool Name: LAN
   - Default Gateway: 192.168.1.1
   - DNS Server: 8.8.8.8
   - Start IP Address: 192.168.1.10
   - Subnet Mask: 255.255.255.0
   - Maximum Number of Users: 50
5. Click Add or Save to create the DHCP pool.
6. For each PC, open Desktop → IP Configuration and select DHCP to obtain an IP address automatically.
7. Switch to Simulation Mode and filter for DHCP packets to observe the four stages of the DORA process.

**DHCP DORA Process**

1. **Discover:** The PC broadcasts a DHCP Discover packet to find a DHCP server.
2. **Offer:** The server offers an available IP address to the PC.
3. **Request:** The PC requests the offered IP address.
4. **Acknowledge:** The server confirms the assignment, and the PC receives its network configuration.

**Discover Packet**

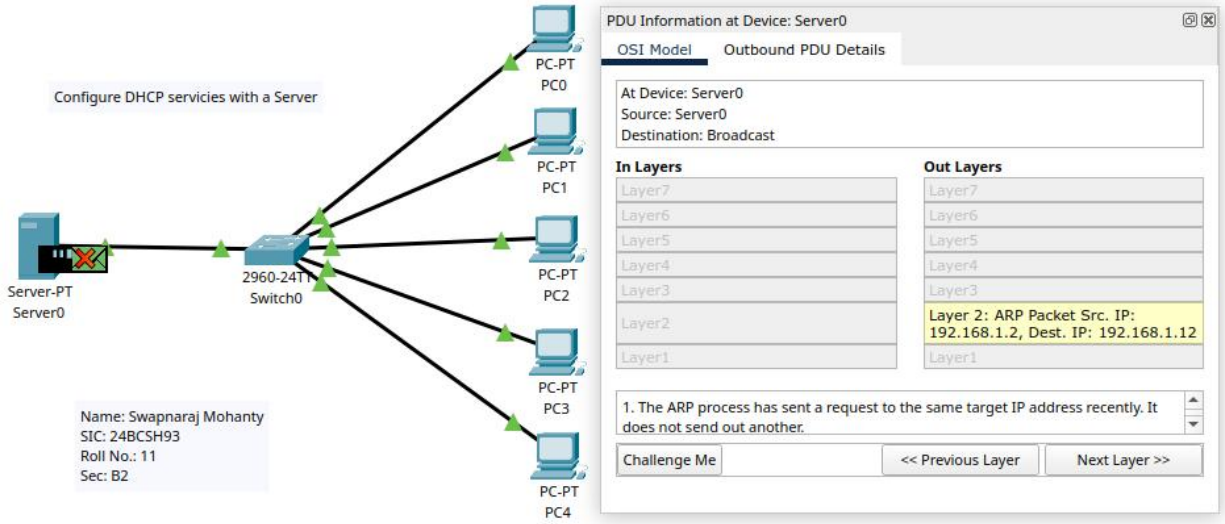




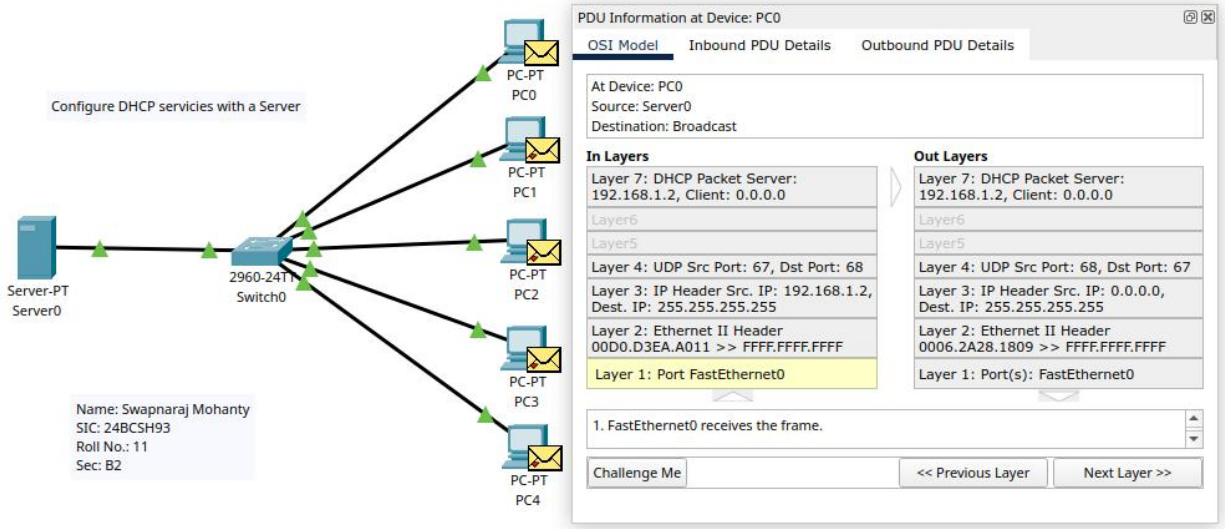
Name: Swapnaraj Mohanty
SIC No: 24BCSH93
Lab Roll No: 11

Date of Experiment: 20th October 2026

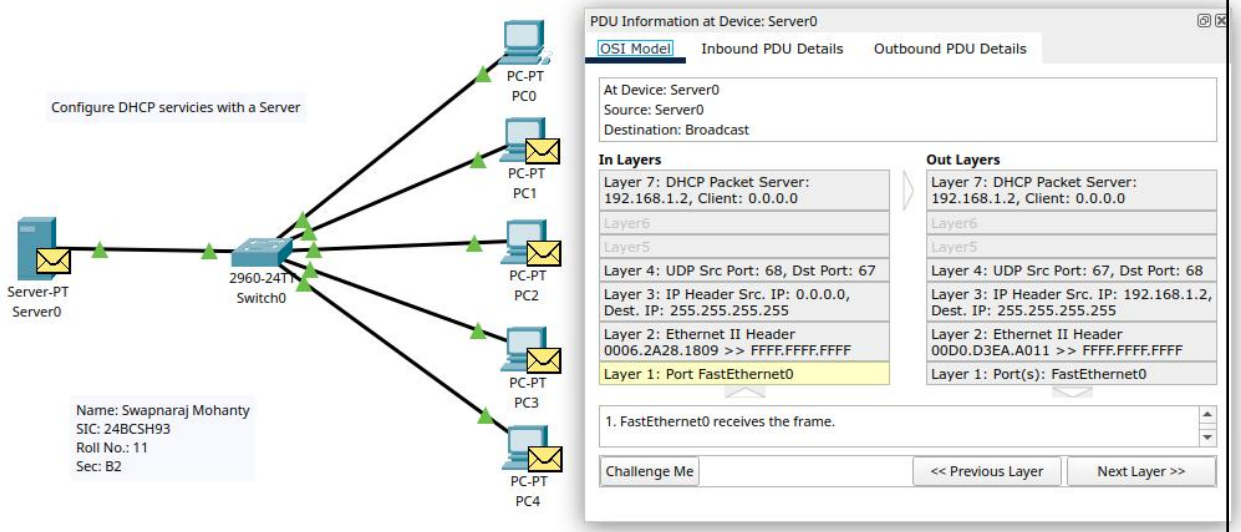
**Offer Packet:**




**Request Packet:**




**Acknowledge Packet:**




Name: Swapnaraj Mohanty
SIC No: 24BCSH93
Lab Roll No: 11
Date of Experiment: 20th October 2026

**Q2. Configure DHCP Services with a Router.**

1. Create one ISR4331 router, one Cisco 2960 switch, and five PCs in Cisco Packet Tracer.

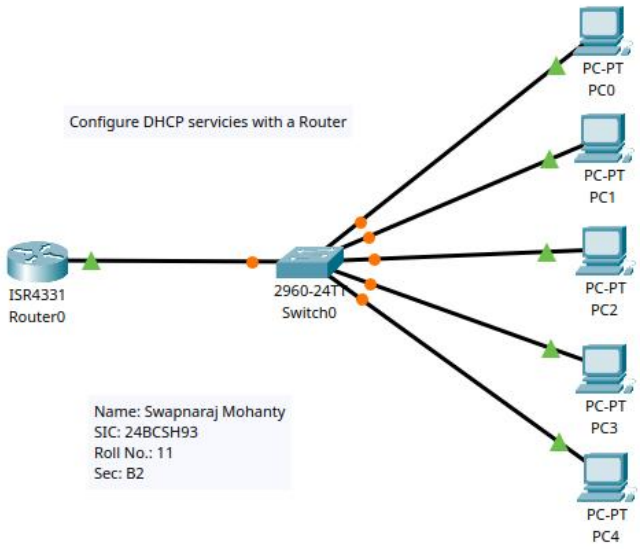
2. Connect Router0 GigabitEthernet0/0/0 to the switch, then connect all five PCs to the switch using Copper Straight-Through cables.

3. Configure the router interface and DHCP service using the following CLI commands:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)#ip dhcp pool MY_LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
```

4. On each PC, open Desktop → IP Configuration and select DHCP to obtain an IP address automatically.
5. Verify that each PC receives a unique IP address starting from 192.168.1.11, with subnet mask 255.255.255.0, default gateway 192.168.1.1, and DNS server 8.8.8.8.
6. Open the PC Command Prompt and execute the following command to test connectivity to the router:

```
C:\>ping
192.168.1.1
```




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)#ip dhcp pool MY_LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
```

Name: Swapnaraj Mohanty
SIC No: 24BCSH93
Lab Roll No: 11
Date of Experiment: 20th October 2026